

● TECHNICAL ARCHITECTURE DOCUMENT

POS System Implementation Plan

Omnichannel Responsive Architecture (Mobile, Tablet, Desktop) with Co-Buy Analytics Certified Excel Export Engine

SYSTEM TYPE	TARGET DEVICES	ANALYTICS INTEGRATION	EXPORT CERTIFICATION
Responsive Web POS / PWA	Cellphone, Tablet, Laptop	Co-Buy Market Basket Engine	100% Co-Buy Tidy Format

1 Executive Summary & Objectives

This document outlines the complete architectural, functional, aesthetic, and data engineering blueprint for a standalone, modern **Point of Sale (POS) System**. The primary objective is to empower cashiers, floor staff, and store managers with a fast, touch-optimized, and aesthetically polished checkout terminal that operates seamlessly across **smartphones, tablets, and desktop/laptop computers**.

Crucially, the system features a dedicated **Co-Buy Analytics Export Engine** that produces automated, standardized Excel reports (`.xlsx` / `.csv`). The exported datasets conform 100% with the transactional schema and noise-cleaning algorithms required by the **Co-Buy Retail Intelligence Platform**, enabling instant drag-and-drop Market Basket Analysis, Association Mining (Apriori), Cross-Selling, and Bundle Recommendations with zero manual data scrubbing.



Core Strategic Goal: Zero-Friction Handshake with Co-Buy

Sales recorded at the counter or floor via phone/tablet/laptop are automatically formatted to the exact schema that Co-Buy expects (`InvoiceNo` , `Product_Name` , `Category` , `Quantity` , `Unit_Price` , `Date`), creating a seamless pipeline from POS checkout to AI-driven retail insights.

2 Multi-Device Responsive UI/UX Architecture

The POS UI utilizes an adaptive layout system that shifts dynamically based on viewport dimensions and input capabilities (touch vs mouse/keyboard):

MOBILE PHONE (360PX - 480PX)

Single-Hand Floor Terminal

Portrait Mode • Handheld & Queue Busting

- ✓ Thumb-friendly single column navigation.
- ✓ Sticky bottom cart drawer showing item count & total.
- ✓ Integrated camera barcode scanner via WebRTC.
- ✓ Full-screen slide-up modal for payment tender & change calculator.
- ✓ Min touch targets: 48px × 48px to eliminate mis-taps.

TABLET (768PX - 1024PX)

Counter-Top Touch Register

Landscape & Portrait • iPad / Android Tablets

- ✓ Adaptive 60/40 split-screen layout.
- ✓ Left 60%: Responsive visual product tile grid with category pills.
- ✓ Right 40%: Persistent live cart with swipe-to-delete and quick qty steppers.
- ✓ Dedicated on-screen number pad for quick price override & tender.
- ✓ Customer-facing display support (dual screen or mirror).

LAPTOP / PC (1280PX - 1920PX+)

Full Workstation Command Center

High-density Desktop • Heavy Barcode Scanning

- ✓ 3-column layout: Category drawer, Product catalog matrix, Live cart.
- ✓ Keyboard shortcuts (**F1** Search, **F2** Discount, **Space** Pay, **Enter** Print).
- ✓ Instant autofocus for external USB/Bluetooth 1D/2D laser scanners.
- ✓ Dual thermal printer support (58mm/80mm ESC/POS) + Cash drawer kick.

3 Aesthetic & Design System (Co-Buy Aligned)

The POS design embodies a high-end, clean, and modern aesthetic matching Co-Buy's design language:



Emerald Growth
#10B981
Primary Actions, Tender, Total



Sapphire Data
#0EA5E9
Search, Filters, Export Engine



Midnight Slate
#0F172A
Dark Theme, Headers, Cards



Clean Canvas
#F8FAFC
High-readability Day Mode

- **Dual Theme Architecture:** Seamless toggle between High-Contrast Dark Mode (ideal for dim venues, bars, cafes) and Crisp Light Mode (ideal for retail boutiques and sunny stalls).
- **Visual Feedback:** Tactile micro-animations on item tap (slight bounce + sound chime), ripple effects, and distinct status color codes (Paid, Pending, Refunded).

4 The Co-Buy Excel Export Engine Specification

Co-Buy's data ingestion pipeline (`/api/upload` & `parse_df_to_transactions`) strictly relies on a clean, tidy, vertical transaction matrix. The POS export module must generate an Excel (`.xlsx`) file that meets these exact column keyword mappings:

Column Header	Requirement	Data Type / Format	Co-Buy Ingestion Keyword Match	Validation & Anti-Noise Rule
InvoiceNo	MANDATORY	String (e.g. POS-20261004-001)	Matches: invoice, order, receipt, trans	Group identifier. Multiple items in one cart share the exact same InvoiceNo.
Product_Name	MANDATORY	Clean String (Title Case)	Matches: product, item, desc, description, name	Must NOT be numeric barcode alone. Exclude noise tokens (discount, refund, void).
Category	MANDATORY	String (Department/Group)	Matches: category, department, dept, group	Used by Co-Buy for Category Association Mining & Department Sales Distribution.
Quantity	MANDATORY	Positive Integer (≥ 1)	Matches: qty, quantity, count, units	Zero and negative values are automatically rejected as invalid noise.
Unit_Price	MANDATORY	Decimal Number (≥ 0.00)	Matches: price, unit_price, cost, rate	Regular unit retail price before transaction-level discount.
Date	MANDATORY	ISO Format (YYYY-MM-DD)	Matches: date, time, datetime, trans_date	Standard ISO 8601 date. Timestamp (HH:MM:SS) optional.
Payment_Method	ENRICHMENT	String (Cash, GCash, Card)	Stored as metadata attribute	Useful for cashier reconciliation and customer payment behavior analysis.
Cashier_Name	ENRICHMENT	String (Staff Name / ID)	Audit trail logging	Enables shift-level productivity and sales attribution tracking.

Exact Structure of Exported Excel Sheet (`CoBuy_Transactions`):

InvoiceNo	Date	Product_Name	Category	Quantity	Unit_Price	Total_Amount	Payment_Method
POS-20261004-101	2026-10-04	Artisan Sourdough Loaf	Bakery	1	145.00	145.00	GCash
POS-20261004-101	2026-10-04	Organic Salted Butter	Dairy	1	180.00	180.00	GCash
POS-20261004-101	2026-10-04	Arabica Cold Brew 250ml	Beverages	2	120.00	240.00	GCash

InvoiceNo	Date	Product_Name	Category	Quantity	Unit_Price	Total_Amount	Payment_Method
P0S-20261004-102	2026-10-04	Almond Croissant	Bakery	2	95.00	190.00	Cash
P0S-20261004-102	2026-10-04	Iced Spanish Latte	Beverages	1	150.00	150.00	Cash



Why This Table Structure Works Instantly in Co-Buy:

Notice how transaction P0S-20261004-101 has 3 distinct rows sharing the same InvoiceNo. Co-Buy's backend parser (app.py) automatically groups these 3 rows into a single market basket: ['Artisan Sourdough Loaf', 'Organic Salted Butter', 'Arabica Cold Brew 250mL'], calculates the basket value (₱565.00), and computes association confidence, lift, and support with zero manual adjustment.

5 Functional Modules Breakdown



1. Item Catalog & Fast Search

Categorized product cards with images, prices, and stock indicators. Real-time instant search by SKU, barcode, or name. Barcode scanner listener for instant 1-scan addition.



2. High-Speed Cart Engine

Add, increment, decrement, and swipe-to-delete. Support for line-item notes, customer discount tags (Senior Citizen, PWD, VIP), and tax toggle (inclusive/exclusive VAT).



3. Multi-Tender Payment

Cash tender with rapid denomination buttons (₱100, ₱500, ₱1000, Exact) and automatic change computation. Support for E-Wallets (GCash QR, Maya), Credit/Debit Cards, and Split Tender.



4. Receipt & Hardware Support

Instant receipt preview with thermal print commands (58mm / 80mm ESC/POS via Web Bluetooth or USB). Digital receipt export via SMS, QR Code, or Email for paperless stores.



5. Shift & Cash Drawer Control

Cashier clock-in with opening cash float. Mid-shift paid in/out tracking. End-of-shift X-Reading (interim tally) and Z-Reading (shift close and cash drawer reconciliation).



6. Offline-First Resilience (PWA)

Service Workers and local IndexedDB database buffer all sales locally. If store internet drops, checkout never freezes. Background sync uploads queue when online.

6 Recommended Technical Stack

To ensure the POS runs effortlessly on low-powered mobile phones while retaining enterprise-grade reliability on laptops, the following modern architecture is recommended:

Layer	Technology	Strategic Rationale
Frontend Client	React 19 / Vite + Vanilla CSS / CSS Modules	Ultra-fast initial load (<1s), 60fps animations, tiny bundle footprint, no heavy framework overhead on mobile browsers.
Offline Storage	IndexedDB (via Dexie.js or idb-keyval)	Stores 100,000+ products and local transactions safely in the browser cache. Enables full offline POS functionality.
Barcode Scanner	HTML5-QRCode / ZXing-JS	Uses smartphone camera directly without requiring native mobile app compilation. Supports USB hardware wedge scanners out of the box.
Excel Export Engine	ExcelJS / SheetJS (xlsx)	Client-side multi-sheet Excel generator. Outputs <code>.xlsx</code> directly on the device with zero server latency or data usage.
Backend API (Optional/Sync)	Lightweight Node.js (Express) or Python (FastAPI/Flask)	Handles centralized inventory sync, multi-terminal shift consolidation, and user access control (Cashier vs Manager).
Database	PostgreSQL or SQLite / Supabase	ACID-compliant storage for sales receipts, product master data, and cashier shift logs.

7 Phased Implementation Roadmap

- PHASE 1 • FOUNDATIONS & DATA MODELING (WEEK 1)**

Schema Architecture & Co-Buy Export Generator

Establish database models (Products, Categories, Orders, OrderItems). Build and unit-test the client-side Excel export generator against Co-Buy's `/api/upload` parser to guarantee 100% upload compliance.
- PHASE 2 • RESPONSIVE FRONTEND & ERGONOMICS (WEEK 2)**

Adaptive Layouts: Mobile, Tablet & Desktop Views

Develop the responsive UI shell. Implement the single-hand mobile cart drawer, tablet 60/40 split-view, and desktop 3-column workstation. Integrate camera-based barcode scanning and touch-friendly numpads.
- PHASE 3 • CHECKOUT & MULTI-TENDER WORKFLOWS (WEEK 3)**

Payment Processing & Change Computation

Implement tender modals for Cash (with fast denomination presets), E-wallets (GCash/Maya reference codes), Credit Cards, and Split Payments. Real-time balance and change calculations with audit safety guards.
- PHASE 4 • HARDWARE & RECEIPTS (WEEK 4)**

Thermal Printing & Offline-First PWA

Configure Web Bluetooth / WebUSB ESC/POS receipt printing (58mm/80mm). Implement Progressive Web App (PWA) service workers with IndexedDB offline queueing and automatic sync on reconnection.
- PHASE 5 • SHIFT MANAGEMENT & QA CERTIFICATION (WEEK 5)**

Cashier Reconciliation, Security & Launch

Implement opening/closing shifts (X and Z readings). Conduct multi-device stress testing on Android phones, iPhones, iPads, and Windows laptops. Verify end-to-end export file import in Co-Buy.

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Acceptance & Quality Criteria

- **Co-Buy Zero-Error Ingestion:** Any exported `.xlsx` file from the POS must pass Co-Buy's file validation without trigger of empty dataset warnings or unparseable columns.
- **Cross-Device Touch Latency:** Adding an item to cart or switching categories must render in $\leq 50\text{ms}$ on mobile devices.
- **Offline Survivability:** Pulling the network cable or turning on Airplane Mode must not interrupt active transactions; transactions must sync cleanly once connection restores.
- **Tender Accuracy:** Zero floating-point calculation errors on sales totals, discounts, or change amounts (using integer cent/centavo arithmetic).

Prepared For: Standalone POS Implementation Project

Target Analytics Platform: Co-Buy Retail Intelligence

Document Status: Complete Implementation Plan v1.0